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Project Title:	Gap Junctions and Microgravity Induced Bone Loss		
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Program/Discipline--Element/Subdiscipline:			
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Human Research Program Risks:	None		
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Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Task Description:

Astronauts exposed to extended periods of unloading experience significant bone loss. As NASA undertakes an exploration initiative involving long-duration spaceflight and exploration of asteroids and Mars, astronauts will encounter prolonged periods of unloading followed by subsequent extra-vehicular activities in a gravity environment. Loss of bone mass and strength will complicate these activities and impair mission efficiency. For instance, considering the magnitude of bone loss occurring during spaceflight (0.5-1.5% loss of bone mineral density (BMD)/month), a fall on an already weakened skeletal system could result in fracture. Therefore, development of countermeasures for microgravity-induced bone loss is critical. We previously demonstrated that osteoblast- and osteocyte-specific deficiency in the gap junctional protein connexin 43 (Cx43) is protective against bone loss induced by unloading via hindlimb suspension (HLS). However, the mechanism by which this occurs is unknown. Osteocytes are well situated in bone to detect and communicate mechanical signals to effector cells and therefore play a pivotal role in bone's ability to sense and respond to unloading from exposure to microgravity. Therefore, we will examine the hypothesis that Cx43 deficiency attenuates the response of osteocytes to unloading resulting from simulated microgravity. Additionally, we will examine whether the response of osteocytes exposed to simulated microgravity in vitro can predict bone response to simulated microgravity in vivo. Using an in vivo and in vitro model, we will complete the following two aims.
Aim 1: Examine the response of wild-type and Cx43-deficient (via CRISPR gene editing) murine osteocytes (OCY454) to simulated microgravity in vitro using a High Aspect Ratio Vessel (HARV).

Aim 2: Examine the effect of simulated microgravity in vivo on bones from osteocyte-specific Cx43-deficient mice. Cx43 will be selectively deleted from mice using a Cre recombinase approach similar to what we previously used to delete Cx43 from osteoblasts and osteocytes.

Using well-established experimental paradigms like the HARV apparatus and HLS greatly increases the probability of successfully completing our aims and attaining important, novel findings. Collectively, our infrastructure and expertise, in combination with multiple unique reagents, place us in an unparalleled position to rapidly and significantly advance knowledge pertaining to the mechanisms underlying microgravity-induced bone loss. These aims and the experiments designed therein represent a substantive departure from our current understanding and will reveal novel targets for countermeasures.

Rationale for HRP Directed Research:

The proposed research will establish the mechanism underlying the protective effect of Cx43 against microgravity-induced bone loss. This will help fill gaps in our understanding of gravity-sensing mechanisms. We will also gain insight as to whether in vitro models of microgravity, such as culturing cells in a HARV apparatus, induce similar changes in osteocytic gene expression as do in vivo models such as HLS. From the perspective of microgravity countermeasures, this information will guide the choice of potential therapeutics.
On the ground, concomitant bone loss is frequently experienced by individuals confined to bed for long durations, whether as a result of neurological injury, trauma, or neuromuscular disease (e.g., cerebral palsy and muscular dystrophy). In these patients, and in healthy patients subjected to extended bed rest, a significant loss of bone occurs. In the case of recoverable disease or when ambulation is still possible, this leaves patients at risk for falls and catastrophic fracture. Understanding the mechanisms involved in bone loss during spaceflight, and the development of countermeasure strategies, will provide a framework for the development of preventative treatments for astronauts and the clinical population.

Research Impact/Earth Benefits:

The proposed research is innovative because it provides the first simultaneous examination of effects of simulated microgravity on genetically modified osteocytes in vivo and in vitro. A few studies have examined the effects of simulated microgravity on bone in vivo and in vitro,but no study has compared the effects of simulated microgravity on osteocytes or genetically modified mice in vivo and in vitro in the same study. Osteocytes play a key role in bone’s mechanosensitivity and adaptation to changes in mechanical loading. Furthermore, we are investigating the novel concept that Cx43 deficiency protects against microgravity-induced bone loss through a non-canonical mechanism involving an interaction between Cx43 and the WNT/ß-catenin pathway. Finally, we will use innovative transcriptomic approaches to compare the response of primary osteocytes exposed to simulated microgravity in vivo to osteocytic OCY454 cells exposed to simulated microgravity in vitro. In this proposal we are leveraging our experience and expertise in bone biology (Donahue) and gene expression (Buck). Our team brings forward a collaborative and unique skill set in order to ensure a complete and comprehensive investigation of the response of osteocytic cells to simulated microgravity.

Aim 1: Examine the response of wild-type (WT) and Cx43-deficient (via CRISPR gene editing) murine osteocytes (OCY454) to simulated microgravity in vitro using a high aspect ratio vessel (HARV).
We have made considerable progress in vitro experiments for aim 1. In this aim we hypothesized that Ocy454 wildtype (WT) cells, which display characteristics of mature osteocytes, exposed to simulated µg display decreased expression of bone anabolic genes and increased expression of bone catabolic genes. Furthermore, this effect would be attenuated in Ocy454 Cx43 deficient (Cx43 KO) cells.

To test our hypotheses, Ocy454 WT and Cx43 KO (generated by CRISPR) cells were first cultured at 37°C (5% CO2) for 12 days with media changes (aMEM;20% FBS; Gibco) every 3 days. Next, Ocy454 cells were trypsinized and cultured on collagen 1 coated microcarrier beads (Cytodex3™; GE Healthcare) at 3 x 106 cells per 50 mg of beads for 2 days. A small portion of cells were also collected in RIPA buffer for western blot protein analysis with anti-mouse Cx43 antibody (Sigma C6219; 1:8000). Finally, microcarrier beads with Ocy454 cells attached were loaded onto disks (n = 3 per group/genotype) and placed in a rotary cell culture system (RCCS; Synthecon). Disks were rotated counterclockwise at 15-17 rpm for 24 hrs to simulate µg by average gravity vector cancellation [4]. Non-rotating (static) disks served as controls. Immediately following the experiment, Ocy454 cells were lysed directly off the beads and mRNA was collected and purified using a RNeasy mini kit (Qiagen). SYBR green based qPCR was run with the average cycle number (Cq) of ß-actin and GAPDH genes used for delta CT normalization.

Cx43 protein (43 kDa) was present in cellular protein fractions from Ocy454 WT cells but not Cx43 KO cells. This confirmed that Ocy454 cells express Cx43, which has not been reported in the literature. Furthermore, this confirmed Cx43 deficiency in our Cx43 KO cells. Following 24 hrs of exposure to simulated µg, one disk per group/genotype showed evidence of contamination and wasn’t included in the analysis. Remaining WT Ocy454 cells showed a nearly 8 fold increase in Ptgs2, encodes Cyclooxygenase-2, and a 3 fold increase in Tnfsf11, encodes RANKL, expression versus

Task Progress:	static controls. Upregulation of these same genes were attenuated in Cx43 KO Ocy454 cells exposed to µg versus static controls. Furthermore, Tnfrsf11B, encodes OPG, an inhibitor of osteoclast formation, was upregulated in Ocy454 KO cells exposed to µg versus static controls. All cells regardless of genotype, when exposed to ug, showed increased Pdpn expression, encodes Podoplanin, a marker of dendrite formation, versus static controls. These results support our hypothesis that osteocytic Cx43's protective effects during unloading may be due to a reduction in genes related to bone resorption. This is in line with in vivo studies that show that osteoblast/osteocyte-specific Cx43 deficiency attenuates key mechanoresponsive genes involved in bone resorption (RANKL/OPG). Conclusions: Osteocytic RANKL/OPG signaling attenuation due to Cx43 deficiency may be one osteoprotective mechanism during unloading. Future experimental studies should confirm these gene expression results and also investigate the role of µg and Cx43 deficiency in altering osteocytic WNT/ß-catenin signaling. Aim 2: Examine the effect of simulated microgravity in vivo on bones from osteocyte-specific Cx43-deficient mice. In Aim 2 we hypothesized that the loss of Cx43 in osteocytes would be protective against microgravity-induced bone loss in vivo. We used a mouse model with Cx43 deleted only in osteocytes and subjected mice to unloading via hindlimb suspension (HLS) to investigate the role of osteocytic Cx43 in bone's response to unloading. Cx43-deficient mice were bred using a DMP1-Cre-Lox system on a C57Bl/6J background. At 6 months old, male wild-type and osteocyte specific Cx43-deficient mice were randomly assigned to control (normal ambulation) or HLS groups for 3 weeks (n=5-8 per group). Bone geometry of the left femur was assessed by micro-CT. Mechanical properties were assessed by 3-point bending to failure. Left quadriceps and gastrocnemius muscles were weighed to assess muscle atrophy. Two-Way ANOVA with Tukey's tests were used to test for significant differences between genotype and treatment ($p<0.05$). Cx43 deficiency in osteocytes led to decreased cortical area fraction (CtAr/TAr) and increased marrow area, periosteal perimeter, and endosteal perimeter. Differences in CtAr/TAr did not remain after HLS as more bone was lost from wild-type than Cx43-deficient mice. Differences in bone parameters and response to unloading were not seen in metaphyseal trabecular bone. Cx43-deficient bones exposed to HLS had greater ultimate load than wild-type HLS, but there were no differences in ultimate stress. No differences were found in muscle mass of the left quadriceps or gastrocnemius muscles of HLS mice relative to control. Conclusions: These results are similar to what was seen in mice with Cx43 deleted in osteoblasts and osteocytes. This suggests that the role of Cx43 in bone's mechanosensing mechanism is mainly found in osteocytes. Cx43-deficient bones display early development of osteopenia and are less susceptible to the negative effects of unloading. However, Cx43 deficiency did not protect against HLS-induced muscle loss. Future work will evaluate the responses of other genes involved in sensing and transducing mechanical loading in Cx43-deficient mice. Reference 4. Hammond, T.G and Hammond, J.M. Am J Physiol Renal Physiol, 2001. 281(1): p. F12-25.
Bibliography Type:	Description: (Last Updated: 01/06/2026)
Articles in Peer-reviewed Journals	Friedman MA, Zhang Y, Wayne JS, Farber CR, Donahue HJ. "Single limb immobilization model for bone loss from unloading." J Biomech. 2019 Jan 23;83:181-9. https://doi.org/10.1016/j.jbiomech.2018.11.049 ; PubMed PMID: 30551918 , Jan-2019